Project Design Phase-II

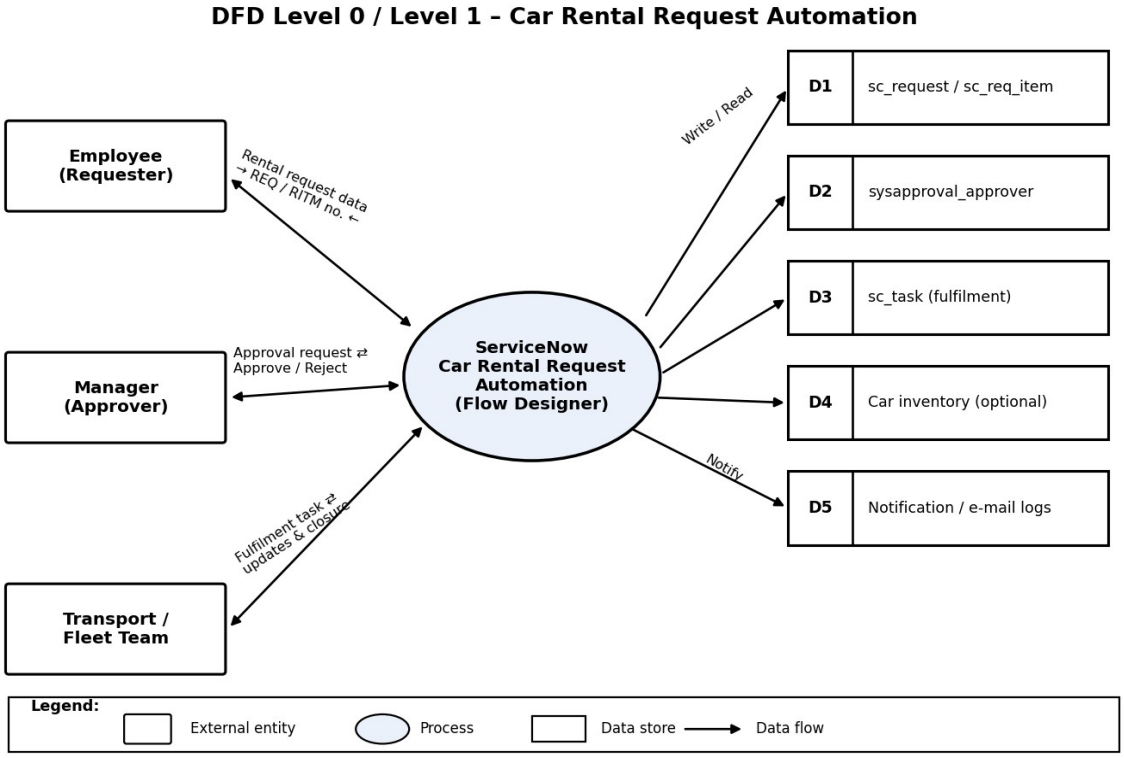
Data Flow Diagrams & User Stories

Date	Team ID	Project Name	Maximum Marks
7 October 2026		Car Rental Request Automation in ServiceNow	4 Marks

Data Flow Diagrams

A Data Flow Diagram (DFD) shows how information moves through a system: where data comes in and goes out, which processes transform it, and where it is stored.

DFD Level 0 / Level 1 – Car Rental Request Automation



External Entities and Main Data Flows

- Employee (requester) sends the car rental request data: dates, locations, car type and reason.
- Manager is sent an approval request and sends back an Approve or Reject decision.
- Transport / Fleet team is sent a fulfillment task and returns the car assignment and task status.
- The ServiceNow platform (central process) creates the REQ, RITM, SCTASK and approval records and triggers notifications.
- The notification engine delivers email and portal messages to everyone involved.

Core Data Stores

- D1: sc_request / sc_req_item (REQ and RITM) – request header and item details
- D2: sysapproval_approver – approval records and decisions
- D3: sc_task – fulfillment tasks for the Transport team

- D4: custom Car Inventory table (optional) – vehicles, their status and assignment history
- D5: sys_email / notification logs – record of all communication

DFD Level 1 – Major Processes

1. P1 – Submit catalog request: the employee completes the variables, creating a REQ and RITM.
2. P2 – Route for approval: Flow Designer raises an approval record and the manager decides.
3. P3 – Create fulfillment task: once approved, an SCTASK is assigned to the Transport group.
4. P4 – Fulfill and close: Transport updates the task, the RITM and REQ close, and notifications go out.
5. P5 – Notify stakeholders: sent at every state change (submission, approval, rejection, closure).

User Stories

User stories for the Car Rental Request Automation product.

User type	Epic	USN	User story / task	Acceptance criteria	Priority	Release
Employee	Catalog request	USN-1	As an employee, I can open the "Car Rental Request" catalog item and submit a request with travel date, pick-up and drop-off locations, duration, car type and reason.	A REQ and RITM are created and a request number is displayed.	High	Sprint-1
		USN-2	As an employee, I get an email or portal notification confirming that my request was submitted.	Notification arrives with the request number and a summary.	High	Sprint-1
	Status tracking	USN-3	As an employee, I can check the current status of my request in My Requests or the Service Portal.	Status and activity stream are shown and kept current.	High	Sprint-1
		USN-4	As an employee, I am notified when my request is approved, rejected or completed.	Correct message is sent, with the reason if rejected or next steps if approved.	High	Sprint-1
Manager	Approval	USN-5	As a manager, I am notified of a pending request and can approve or reject it, adding comments if I wish.	Approval record is updated; the request moves forward or is cancelled.	High	Sprint-1
		USN-6	As a manager, I can see the main request details (dates, locations, car type, reason) on the approval form.	All relevant variables appear on the approval screen.	Medium	Sprint-1
Transport team	Fulfillment	USN-7	As a Transport team member, I automatically get an SCTASK when a request is approved.	Task shows in the group queue with correct assignment and variables.	High	Sprint-1
		USN-8	As a Transport team member, I can update the task (assign a car, add work notes) and close it, which closes the parent request.	Closing the task updates the RITM / REQ state and the requester is notified.	High	Sprint-1
Admin	Configuration & reporting	USN-9	As an admin, I can capture every configuration change in an Update Set for promotion.	Update Set holds all relevant records and can be committed on the target instance.	High	Sprint-1
		USN-10	As an admin, I can view reports and dashboards of open requests, cycle times and SLA performance.	Reports can be filtered by department, car type and status, and the data is accurate.	Medium	Sprint-2

Note: every user story maps to a Flow Designer stage, Business Rule or catalog configuration, and is implemented following ServiceNow good practice (Update Sets, ACLs, Flow error handling).